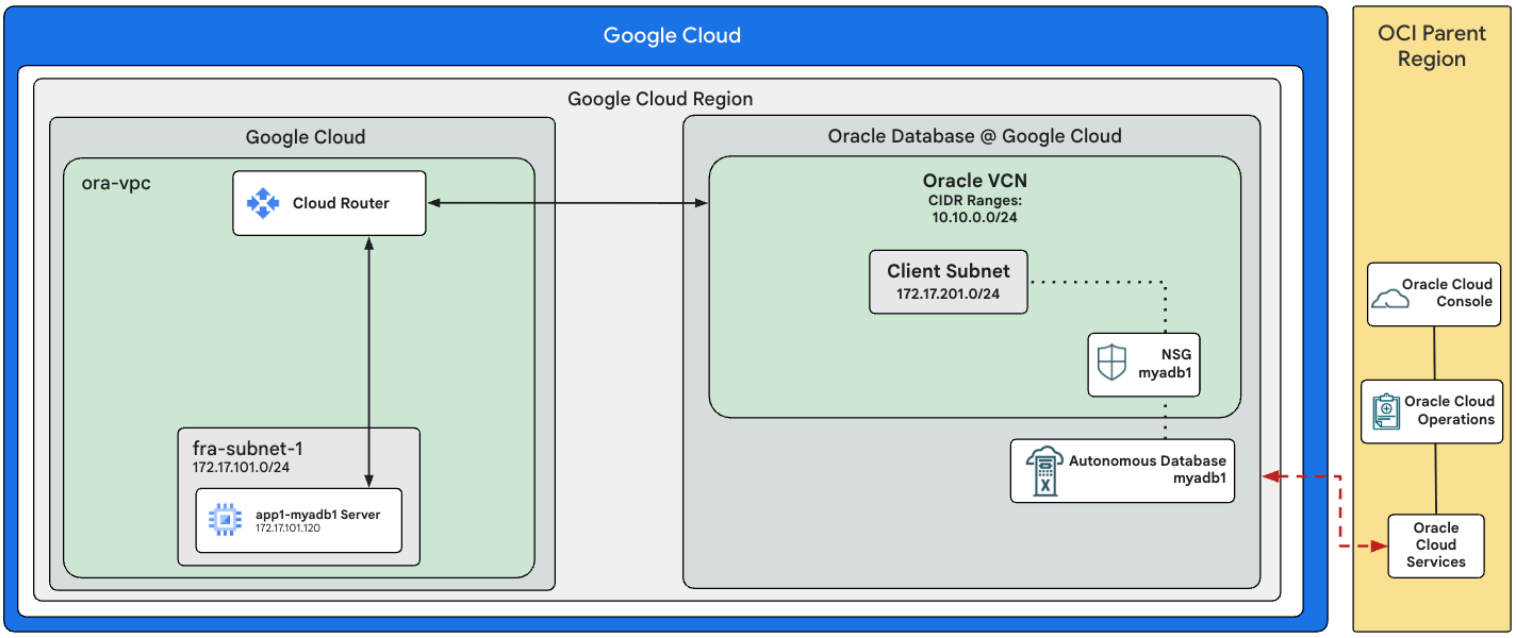
Design and Manage Oracle Workloads on Google Cloud + Vertex AI Coffee Demo **Hands-On Lab Guide**

Introduction

This exercise shows how to create an Autonomous Database instance within Oracle Database@Google Cloud. It walks through 3 steps:

1. Create the Autonomous Database
2. Create a client VM in Google Compute Engine
3. Connect to the client VM and import a sample dataset

The final state of your environment will include an application VM running in Google Compute Engine and an Autonomous Database running in Oracle Database@Google Cloud.



Create an Autonomous Database

Autonomous Database creation automates several tasks for the database administrator. Connectivity between the Oracle-managed network and the Google VPC network is automatically configured, along with a Cloud DNS forwarding zone to allow database clients to connect to the database by the private endpoint's hostname.

You have the option to create an Autonomous Database instance through the Google cloud console or with the `gcloud` command. Oracle Database@Google Cloud also supports Terraform to create and destroy Autonomous Database Instances, although that is out of scope for this exercise.

Option 1: Console-based creation

You will create an Autonomous Database within the `<My Project ID>` GCP project. Use your LDAP for the instance name. For example, joesmith@google.com would use `joesmith` for your instance name

1. Open the Google Cloud console
2. Select the project `<My Project ID>`
3. Navigate to the Oracle Database@Google Cloud page and list the Autonomous Databases
4. Click `Create`
5. In the **Instance details** section, complete the following:
    a. Instance ID: Your LDAP
    b. Database name: Your LDAP
    c. Database display name: Your LDAP
    d. Region: `europe-west2`
6. In the **Workload configuration** section, choose `Transaction Processing` as the workload type
7. In the **Database configuration** section, complete the following:
    a. License type: `License Included`
    b. Database version: `23ai`
    c. ECPU count: `2`
    d. Select `Enable compute autoscaling`
    e. Storage (TB): `0.02`
    f. Select `Enable storage autoscaling`
8. In the **backup retention period** section, enter `1` day
9. In the **Administrator credentials** section, enter a password such as `My53cur3Pa55w0rd`
10. In the **Networking** section, complete the following:
    a. Select `Private endpoint access only`
        i. From the **Associated network** drop-down, select `ora-vpc`
        ii. In the Subnet Range field, enter `172.17.201.0/24`
        iii. Click on **Advanced network settings**
            1. In the **Hostname prefix** field enter your LDAP
        iv. Leave **Allow public access** and **Require mutual TLS (mTLS) authentication** unchecked
11. In the **Operational notifications and announcements contacts** section, enter your email address
12. Click **Create**

The operation will take approximately 15 minutes to complete.

Note: If you receive the following error message while running any gcloud command, please run the authentication command and follow the instructions

You do not currently have an active account selected.

Please run:

$ gcloud auth login

to obtain new credentials.

Option 2: Create from Cloud Shell

You will create an Autonomous Database within the `<My Project ID>` GCP project. Use your LDAP for the instance name. For example, joesmith@google.com would use `joesmith` for your instance name

1. Open the Google Cloud console
2. Select the project `<My Project ID>`
3. Navigate to the Oracle Database@Google Cloud page and list the Autonomous Databases
4. Open Cloud Shell
5. Use the following command to create an Autonomous Database. Modify the `ADB_ID` variable to match your LDAP.

```shell
export ADB_ID=<MY_LDAP>
export PROJECT_ID=<My Project ID>
export GCP_REGION=europe-west2
export VPC_NETWORK_ID=ora-vpc
export CIDR=172.17.201.0/24
export ADB_PW=My53cur3Pa55w0rd

gcloud oracle-database autonomous-databases create $ADB_ID \
--project=$PROJECT_ID \
--location=$GCP_REGION \
--database=$ADB_ID \
--display-name=$ADB_ID \
--network=projects/$PROJECT_ID/global/networks/$VPC_NETWORK_ID \
--cidr=$CIDR \
--admin-password=$ADB_PW \
--properties-license-type=LICENSE_INCLUDED \
--properties-compute-count=2 \
--properties-db-version=23ai \
--properties-db-workload=OLTP \
--properties-backup-retention-period-days=1 \
--properties-data-storage-size-gb=20 \
--properties-is-storage-auto-scaling-enabled \
--async
```

The above command includes the `--async` switch. This switch runs the operation in the background, reducing the risk of a network interruption causing a failure of the operation. The Autonomous Database will be visible in the Google Cloud console once the database provisioning process has begun.

View Your Autonomous Database

Your Autonomous Database instance should be visible in the Google Cloud console.

1. Open the Google Cloud console
2. Select the project `<My Project ID>`
3. Navigate to the Oracle Database@Google Cloud page and list the Autonomous Databases by clicking the link or searching for **Oracle Database@Google Cloud** in the search bar at the top.
4. Click on the Autonomous Database ID that matches the instance you created in the previous step.
5. View the details of your Autonomous Database
    a. View the **Database version** and **ECPU count**
    b. View the **Client subnet** and **Private endpoint URL** to see the hostname that you can use to connect to the database.
    c. Click on the **Connections** menu item at the top of the database details page
        i. From this page you can download the **database wallet** and view the **connection strings** for connecting an application
        ii. Note the multiple service names that relate to the different performance characteristics you can use to connect
            1. View the hostname and service names
            2. Mutual TLS connections require the database wallet and use port 1522
            3. Standard TLS connections do not require a wallet and use port 1521

Create a Client VM

** Wait until the Autonomous Database instance has been created before attempting this step**

You will now create a client VM that can connect to the Autonomous Database. The below steps include a startup script that will configure the VM based on the hostname. Create the VM using the format `app1-LDAP` (replace LDAP with the name of the database that you created in the previous step).

For example, if the database created in the previous step was named `joesmith` the VM would be named `app1-joesmith`.

The startup script is configured to perform 5 steps:

1. Download the required operating system packages to support the Oracle 23ai instant client
2. Create an oracle user in the operating system
3. Download and unzip the Oracle 23ai instant client
4. Determine the expected database name based on the hostname of the VM instance
5. Query the Oracle Database@Google Cloud API for the connection string details and insert those into the client's `tnsnames.ora` file.

You can choose to create the client VM through the Google Cloud console or via the `gcloud` command line in Cloud Shell.

Option 1: Console-based creation

In this step you will create a Google Compute Engine VM instance that will be used to communicate with the Autonomous Database.

1. Open the Google Cloud console
2. Select the project `<My Project ID>`
3. Navigate to the Compute Engine page and view any existing VMs
4. Click `Create Instance`
5. In the **Machine configuration** section, complete the following:
    a. Name: `app1-LDAP` (replace LDAP with your LDAP)
    b. Region: `europe-west2`
    c. Zone: `europe-west2-b`
    d. Machine type: `n2-standard-2`
6. In the OS and storage section, complete the following:
    a. Click `Change` to select the machine image
        i. Operating system: `Ubuntu`
        ii. Version: `Ubuntu 24.04 LTS x86/64, amd64` (NOT the "Minimal" version)
        iii. Boot disk type: `Standard persistent disk`
        iv. Size: `10GB`
    b. Click `Select` to confirm
7. Do not change any fields in the **Data protection** section
8. In the **Networking** section, complete the following:
    a. In the **Network interfaces** section, edit network interface 1 and complete the following:
        i. Network: `ora-vpc`
        ii. Subnetwork: `lon-subnet-1`
    b. Click Done to confirm
9. Do not change any fields in the **Observability** section
10. In the **Security** section, complete the following:
    a. In the **Access scopes** section, select `Allow full access to all Cloud APIs`
11. In the **Advanced** section, complete the following:
    a. In the **Metadata** section, click `Add item` and complete the following:
        i. Key 1: `startup-script-url`
        ii. Value 1: `https://storage.googleapis.com/oracle-partner-demo-bucket/startup-scripts/ubuntu-oracle-startup-script.sh`
12. Click `Create` to launch the VM

Option 2: Create from Cloud Shell

In this step you will create a Google Compute Engine VM instance that will be used to communicate with the Autonomous Database.

1. Open the Google Cloud console
2. Select the project `<My Project ID>`
3. Navigate to the Compute Engine page and view any existing VMs

4. Open Cloud Shell
5. Use the following command to create an Autonomous Database. Modify the `VM_INSTANCE_ID` variable to match your LDAP.

```shell
export VM_INSTANCE_ID=app1-<LDAP>
export PROJECT_ID=<My Project ID>
export GCP_REGION=europe-west2
export VPC_NETWORK_ID=ora-vpc
export VPC_SUBNET_ID=lon-subnet-1

gcloud compute instances create $VM_INSTANCE_ID\
  --project=$PROJECT_ID \
  --zone=$GCP_REGION-b \
  --machine-type=n2-standard-2 \
  --network-interface=network-tier=PREMIUM,stack-type=IPV4_ONLY,subnet=$VPC_SUBNET_ID \
  --maintenance-policy=MIGRATE \
  --provisioning-model=STANDARD \
  --scopes=https://www.googleapis.com/auth/cloud-platform \

--metadata=startup-script-url=https://storage.googleapis.com/oracle-partner-demo-bucket/startup-sc
ripts/ubuntu-oracle-startup-script.sh \
  --tags=ssh \

--create-disk=auto-delete=yes,boot=yes,device-name=$VM_INSTANCE_ID-boot,image-project=projects/ubu
ntu-os-cloud,image-family=ubuntu-2404-lts-amd64,mode=rw,size=10,type=pd-standard \
  --no-shielded-secure-boot \
  --shielded-vtpm \
  --shielded-integrity-monitoring \
  --labels=goog-ec-src=vm_add-gcloud \
  --reservation-affinity=any
```

Connect to the Client and Import Data

Task 1: Log in to Client VM

In this step you will log in to the Google Compute Engine VM instance that was created in the previous step and import data.

1. Open the Google Cloud console
2. Select the project `<My Project ID>`
3. Navigate to the Compute Engine page and view any existing VMs
4. Click on the VM that you created in the previous step
5. Click SSH
    a. Wait for the window to connect. If you receive an error message, click the `Retry without Cloud Identity Aware Proxy` link and the client should connect

Task 2: Inspect the Database with gcloud

You can use the gcloud command to query details of your Autonomous Database instance:

```shell
gcloud auth login


gcloud oracle-database autonomous-databases list \
  --location=europe-west2 \
  --format="table[box,all-box,title='Databases'] \
  (database, \
  properties.state, \
  properties.dbVersion, \
  properties.dbWorkload, \
  properties.privateEndpointIp, \
  properties.privateEndpoint)"
```

The above command will display a table of Autonomous Database instances and list the name, workload type, version, status, and private endpoint details. An example of this output from a different region is below:

```
┌──────────────
```

The VM instance's startup script should download and configure the Oracle 23ai instant client and configure a `tnsnames.ora` file using the connection details for your Autonomous Database instance. You can verify this from the SSH session as the Oracle user.

```shell
$ cat $ORACLE_HOME/network/admin/tnsnames.ora

joesmith = (description=
(retry_count=20)(retry_delay=3)(address=(protocol=tcps)(port=1521)(host=joesmith.adb.us-ashburn-1.
oraclecloud.com))(connect_data=(service_name=g24da7e94756f60_joesmith_high.adb.oraclecloud.com))(s
ecurity=(ssl_server_dn_match=no)))
```

The script was successful if you see the name of your Autonomous Database followed by a connection string. Test connectivity to the database by running the `sqlplus` command and entering the password that you used to create the Autonomous Database instance.

```
sqlplus admin/My53cur3Pa55w0rd@ADB_ID
```

Replace `ADB_ID` with the name of your Autonomous Database instance. You will now be at a SQL prompt and can directly query the database:

```none
SQL> select * from dual;

D
-
X
```

You can download the setup script and directly execute it as the default user (which has sudo privilege) in your VM if it fails initially. Also install your favorite editor.

```none
$ curl -O
https://storage.googleapis.com/oracle-partner-demo-bucket/startup-scripts/ubuntu-oracle-startup-sc
ript.sh

$ bash ubuntu-oracle-startup-script.sh

$ sudo apt install vim nano
```

If you get the following error message while logging in the oracle user , fix the last line in the .profile file. There has to be a single .ora_client.env without a dot in the end. After fixing it, reconnect or rerun the profile with **. .profile** command.

```
None
-bash: /home/oracle/.ora_client.env.: No such file or directory
```

Task 4: Import the Dataset

Finally, import the dataset from Google Cloud Storage. One common method for migrating into Autonomous Database is to export your source database, copy the export to Google Cloud Storage, and then import into Autonomous Database. A small dataset has already been exported into Google Cloud Storage for your use.

After logging in, change to the `oracle` user account.

```
None
sudo su - oracle
```

Next, log in to the database using sqlplus and create a credential to access the Google Cloud Storage bucket:

```
SQL
set define off
begin
  DBMS_CLOUD.CREATE_CREDENTIAL(
    credential_name => 'GOOGLE_CRED_NAME',
    username => 'GOOG1ENQ7Y43MZDDAZQJZ7PKU4AWGFUP7GZHQP37L5N7C7A6L2V7QFBT77O35',
    password => 's9ccpWUdG/O3iaurUcUOCa6t4VQWmFargAZCxxKH'
  );
END;
/
```

Leave the `sqlplus` session by typing `exit` and you are ready to import the dump file.

Import the datapump export file using the `impdp` command. Replace `$ADB_ID` in the command with the name of your Autonomous Database instance:

```
Shell
$ impdp admin/My53cur3Pa55w0rd@$ADB_ID \
```

```
    directory=DATA_PUMP_DIR \
    credential=GOOGLE_CRED_NAME \
    tables=PIZZAS,PIZZA_TYPES,ORDERS,ORDER_DETAILS \
    remap_tablespace=USERS:DATA \
    dumpfile=https://storage.googleapis.com/oracle-partner-demo-bucket/pizza.dmp
```

The database has been imported and you can now connect to the database with `sqlplus`. Connect as the admin user and find how many Hawaiian pizzas were sold by month:

```
SQL
$ sqlplus admin/My53cur3Pa55w0rd@$ADB_ID

SELECT
    TRUNC(o."date", 'MM') AS order_month,
    COUNT(DISTINCT o.order_id) AS number_of_hawaiian_orders
FROM
    orders o
JOIN
    order_details od ON o.order_id = od.order_id
JOIN
    pizzas p ON od.pizza_id = p.pizza_id
JOIN
    pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
WHERE
    LOWER(pt.name) LIKE '%hawaiian%'
GROUP BY
    TRUNC(o."date", 'MM')
ORDER BY
    order_month;
```

Optional: Ensure the compute service account has the right privileges

If the organization you are using has certain policies in place, you might need to add these permissions. This is not needed in the Free Trial organizations.

1. Open the Google Cloud console
2. Select the project `<My Project ID>`
3. Navigate to the Service Accounts page (you may need to select the project again)
4. Click on the three vertical dots to the right of compute@developer.gserviceaccount.com
5. Select 'Manage Permissions' -> 'Manage Access'
6. Grant the roles
    a. "Oracle Database @ Google Autonomous Database Viewer"
    b. "Vertex AI User"

Additional Information

The following links provide more information about using Oracle Database@Google Cloud

- Getting Started with Oracle Database@Google Cloud
- Create Autonomous Databases
- Command Reference for gcloud oracle-database autonomous-databases create
- Migrate Data to Oracle Database@Google Cloud using Oracle Data Pump
- Cross-Region Disaster Recovery with Autonomous Database
- Deleting Google Cloud Projects